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TELECOM SYSTEM IOM – NEW RADAR SYSTEM

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REVISION RECORD

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1 PURPOSE AND SCOPE

1.1 SCOPE OF DOCUMENT

This Installation, Operation, and Maintenance (IOM) document defines the technical description, installation requirements, system architecture, operation, and maintenance procedures for the vessel's Radar System comprising the SIMRAD HALO24 Radar Scanner and R3016 16-inch Display Unit.

The document covers the complete radar installation including the radar antenna/scanner, display unit, interconnecting network and power cabling, mounting arrangements, and associated accessories required to provide short and medium-range marine radar surveillance for navigation and situational awareness.

The purpose of this document is to:

- Describe the overall system configuration and equipment layout
- Provide installation and interfacing requirements
- Explain normal operating procedures and user controls
- Define routine inspection, preventive maintenance, and troubleshooting guidelines
- Ensure safe, reliable, and continuous radar operation onboard

The SIMRAD HALO24 radar system provides real-time target detection, tracking, and navigation support by transmitting and receiving radar signals and displaying echoes on the dedicated R3016 16-inch colour monitor. The system is intended for use as a navigation aid to enhance vessel safety during routine operations, restricted visibility, and coastal or offshore navigation.

1.2 SYSTEM DESCRIPTION

The system consists of the following major equipment components:

ITEM	EQUIPMENT DESCRIPTION	MAKE	MODEL	NOS
01	X-Band radar transceiver unit	SIMRAD	HALO24	1
02	Radar Control unit with display	SIMRAD	R3016	1
03	Din Rail Power supply units	Phoenix Contact	QUINT-PS/1AC/24DC/5	2
04	Ethernet switch	Phoenix Contact	1085173	1
05	Junction box	VIVO	OEM	1

Each of the main components of the system is described in more detail below.

1.3 EQUIPMENT DESCRIPTION

1.3.1 X-BAND RADAR TRANSCEIVER UNIT SIMRAD- HALO24

The SIMRAD HALO24 is a high-performance marine radar dome designed to deliver advanced target detection, navigation support, and collision awareness from a single compact unit. The HALO24 radar employs pulse compression radar technology which combines high resolution with long range and excellent close-in detection performance, making it suitable for a wide range of marine navigation scenarios including harbor approaches, coastal waters, offshore transits, and low-visibility conditions.

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Operating within the X-band frequency range, the HALO24 has a maximum range of up to 48 nautical miles and can detect objects from very short distances to long distance targets, enhancing situational awareness and safe vessel operation. The radar rotates at variable speeds up to 60 revolutions per minute (RPM), enabling rapid 360° sweeps and near-real-time updates of radar returns. This fast rotation capability is particularly beneficial for close-range navigation and collision avoidance.

The HALO24 radar incorporates Velocity Track™ Doppler technology, which uses the Doppler effect to determine the relative motion of targets and provides intuitive visual feedback on whether objects are approaching or moving away. This feature improves the operator’s ability to identify potential hazards and make informed navigational decisions without the need to manually select individual targets.

In addition to Velocity Track™, the HALO24 supports advanced radar modes including simultaneous dual-range operation, which allows near and far range scans to be viewed together, and MARPA target tracking (where supported by the connected radar display unit), which can automatically track multiple targets and display their speed and course. These capabilities further enhance collision avoidance and target monitoring.

The HALO24 radar is designed for durability and reliability in harsh marine environments, with IPX6-rated waterproof construction and robust mechanical design. Typical power requirements are suitable for 12 V or 24 V DC vessel electrical systems, and standard network interfaces such as Ethernet allow connection to compatible multifunction displays (MFDs) or external processing units for radar control and display integration.

Overall, the SIMRAD HALO24 radar combines advanced scanning performance, rapid update rates, and intuitive motion sensing technology to provide a versatile and effective radar solution for modern marine navigation and situational awareness.

SIMRAD- HALO24 installed at the following location:

- NAVIGATION DECK- STARBOARD SIDE (86-RR-601)

1.3.2 RADAR CONTROL UNIT WITH DISPLAY SIMRAD R3016

The SIMRAD R3016 is a dedicated radar control and display unit designed to serve as the primary operator interface for compatible SIMRAD radar systems, such as the HALO24 Pulse Compression Radar. It integrates a 16-inch widescreen color display with an embedded control processor and user interface to present radar echoes, settings, and navigational information in an intuitive and accessible format. The R3016 features a 16-inch LED-backlit TFT LCD with a resolution of 1366 × 768, providing a wide and detailed plan-position indicator (PPI) presentation essential for marine radar interpretation. The display supports multiple languages and on-screen menus, allowing operators to adjust radar settings and view target information without obstructing key visual data.

Control inputs are provided through a combination of integrated direct-access menu keys and a rotary dial interface beside the display. These controls enable rapid selection of radar functions, range scales, filters, and tracking features, even under challenging marine conditions or while wearing gloves.

The R3016 is engineered for robust marine use with IPX6 water-resistance and watertight rear connectors, making it suitable for flush-mount installation in a pilothouse console or bracket-mount installation where exposure to spray and moisture may occur.

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Connectivity options include Ethernet for radar data and networking, four NMEA 0183 ports, and NMEA 2000 interface for integration with navigation sensors such as heading, speed, and AIS. HDMI output is provided to support an optional secondary display or to create a remotely operated clone station at a second helm position. The R3016 operates from a standard 12–24 V DC power supply and typically consumes approximately 20 W. The unit is built to operate reliably in marine environments from -15 °C to +55 °C and meets international performance and EMC standards applicable to small craft radar systems.

SIMRAD- R3016 installed at the following location:

- NAVIGATION DECK- WHEELHOUSE (86-RR-603)

1.3.3 DIN RAIL POWER SUPPLY UNIT PHOENIX CONTACT QUINT-PS/1AC/24DC/5

The Phoenix Contact QUINT-PS/1AC/24DC/5 is a high-performance DIN-rail mountable industrial 24 V DC power supply designed to deliver a stable and reliable DC output for demanding control, automation, and communication systems. This power supply belongs to the QUINT POWER series, which is engineered for robust operation in industrial, marine, and offshore environments where continuous and secure power delivery is essential.

The unit accepts a standard single-phase AC input (typically 110–240 V AC) and converts it to a regulated 24 V DC output at a rated current of 5 A, providing up to 120 W of DC power to downstream equipment. The QUINT-PS series incorporates advanced surge suppression, inrush current limiting, and high efficiency design, ensuring minimal interruption even under unstable mains conditions.

Designed to support critical systems such as communication equipment, PLCs, monitoring devices, sensors, and control panels, the QUINT-PS/1AC/24DC/5 offers excellent load regulation, low output ripple, and high immunity to electrical noise. It features built-in diagnostic indicators for power and fault status, facilitating rapid visual checks of operational health.

Safety and reliability are enhanced through multiple protective functions, including:

- Overload protection
- Short-circuit protection
- Overvoltage and overtemperature shutdown

The power supply can withstand harsh industrial environments, with robust construction and thermal management that allows continuous operation at full load across a wide ambient temperature range. Its compact design and standardized DIN-rail mounting system enable straightforward installation in instrument racks, control cabinets, or power distribution panels.

The QUINT-PS/1AC/24DC/5 power supply is widely used to provide dependable 24 V DC power to electronic systems where uninterrupted operation is critical for safety, control, and communications.

Din Rail Power supply unit installed at the following location:

- NAVIGATION DECK- WHEELHOUSE
- A-DECK – MCR- GMDSS Cabinet

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1.3.4 ETHERNET SWITCH PHOENIX CONTACT 1085173

The Phoenix Contact Ethernet Switch, Part No. 1085173, is an industrial-grade unmanaged Ethernet switch designed for reliable data networking in automation, control, and communication applications. Engineered to support robust and uninterrupted data transmission in demanding environments such as marine, offshore, and industrial installations, this device enables multiple Ethernet devices to interconnect over a local area network (LAN) with low latency and high availability.

This Ethernet switch provides a set of 10/100 Mbps Fast Ethernet ports for wired network connections, allowing sensors, controllers, human machine interfaces (HMIs), communication modules, and monitoring devices to share data in real time. As an unmanaged switch, it requires no configuration and supports plug-and-play operation, simplifying system integration and reducing commissioning time.

The unit incorporates industrial hardened design features such as wide operating temperature tolerance, surge protection, and vibration-resistant construction to withstand shipboard and factory floor conditions. The robust housing and DIN-rail mountable form factor allow secure installation within equipment cabinets or control panels. Status LEDs on each port provide visual indication of link presence, activity, and connection speed, supporting quick fault detection and maintenance.

The switch provides the reliable data infrastructure necessary for interconnecting communication equipment, automation devices, and monitoring systems within the vessel's network architecture. It's simple, maintenance-free operation and industrial durability make it ideal for supporting mission-critical network communication where consistent performance is essential.

Ethernet switch installed at the following location:

- NAVIGATION DECK- STARBOARD SIDE -JB (86-RR-604)

2 REFERENCE DRAWINGS

The following reference drawings shall be consulted prior to installation, operation, or maintenance of the equipment.

No	Document No.	Rev	Title or Description
1	WCPOKWTP-450-TE-(VAE)BF-029	1	Telecom Cable Block Diagram - New Radar System
2	WCPOKWTP-450-TE-(VAE)DG-056	1	Telecom Typical Installation Drawing - New Radar System
3	WCPOKWTP-450-TE-(VAE)DG-057	0	Telecom Termination Drawing - New Radar System

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3 MANUFACTURERS' MANUALS

This section includes the manufacturers' manuals for the installation, operation, and maintenance of the equipment.

3.1 INSTALLATION INSTRUCTIONS

3.1.1 INSTALLATION INSTRUCTIONS OF X-BAND RADAR TRANSCEIVER UNIT SIMRAD- HALO24

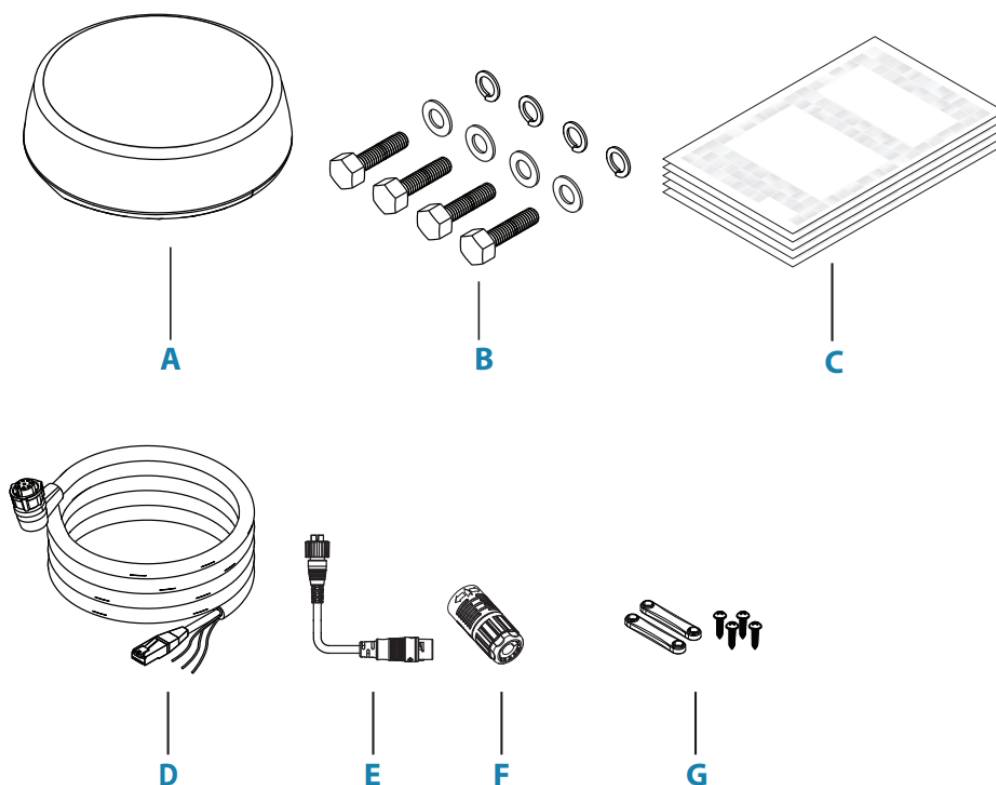


- SPECIFICATIONS

Electrical	
Power Consumption max	29W
Power Consumption typical	20W
Recommended Fuse Rating	5A
Supply Voltage	10.5 - 31.2 VDC
Environment	
Humidity	95% RH at 35°C (95°F)
Operating Temperature Range	-25° to + 55°C
Waterproofing	IPX6
Connectivity	
Ethernet Ports	100 Base T
Radar	
Beam width	3.9° H nominal (-3dB) 2° with target separation on high x 22° V
Max Range, scale	48 NM
Max Wind Speed	100 knots (51 m/s)
Min Range, scale	50 m / 200 ft / 50 yd
Radar Modes	Harbor, Offshore, Weather, Bird, Dual Range, North Up, Course Up, Relative motion, True Motion
Radar Technology	Pulse Compression
Rotation speed	Up to 60 RPM

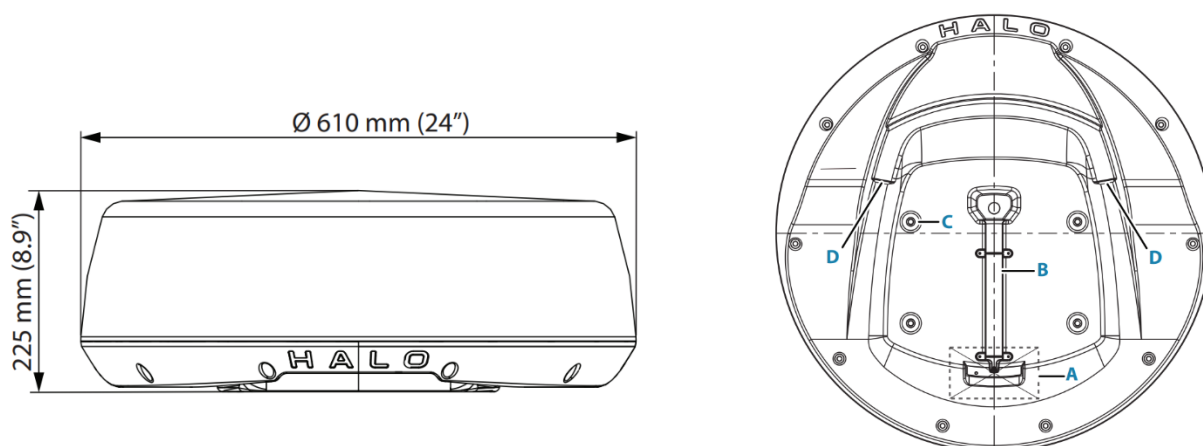
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- WHAT'S IN THE BOX



Key	Description
A	Halo Radar
B	Mounting screws and washers (Screw (M8x30), 4x/Flat washer, 4x/Spring washer, 4x)
C	Documentation pack
D	Interconnection cable
E	Ethernet adapter RJ45 to 5-pin, 1.5m (4.9 ft)
F	Waterproof cable boot for the Ethernet adapter
G	Cable retainer kit (Cable retainer clip, 2x/Screw (M2.5x8), 4x)

- DIMENSIONS



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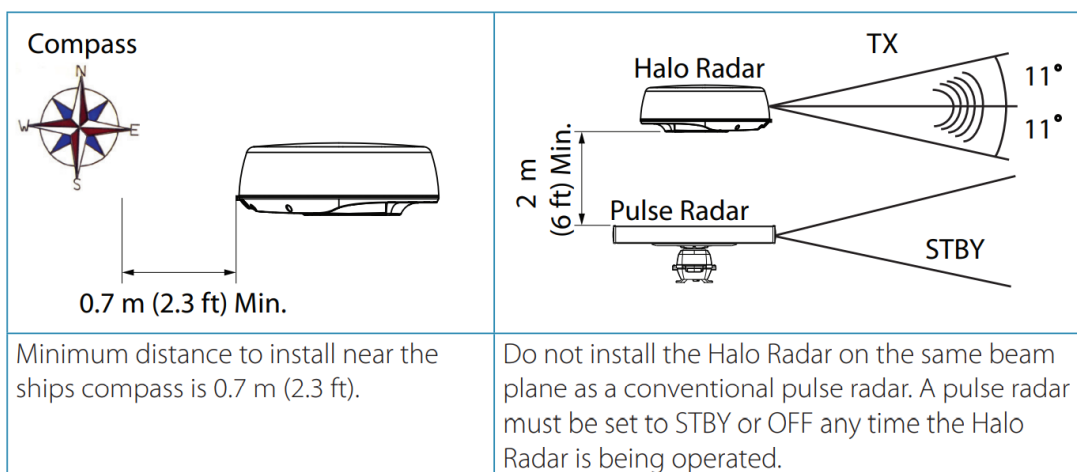
Key	Description
A	Cable entry area
B	Cable retention channel
C	4 Bolt holes M8 x 30 mm
D	LED accent light

- CHOOSE THE SCANNER LOCATION

The performance of the SIMRAD HALO24 radar is significantly influenced by the mounting location of the scanner. For optimum detection and coverage, the radar scanner should be installed at the highest practical position on the vessel, clear of obstructions such as masts, funnels, antennas, or deck structures that may block or reflect the radar beam. Installing the HALO24 at an elevated position improves the radar horizon and increases the maximum detection range, enabling earlier identification of vessels, coastlines, and navigational hazards. However, excessive mounting height may create a larger blind area directly around the vessel, where very close targets may not be detected. Additionally, higher installation heights can slightly reduce the radar's ability to detect small targets within heavy sea clutter. Therefore, the scanner should be mounted at a balanced height that provides maximum coverage while minimizing blind spots and interference, ensuring reliable target detection for safe navigation under all operating conditions.

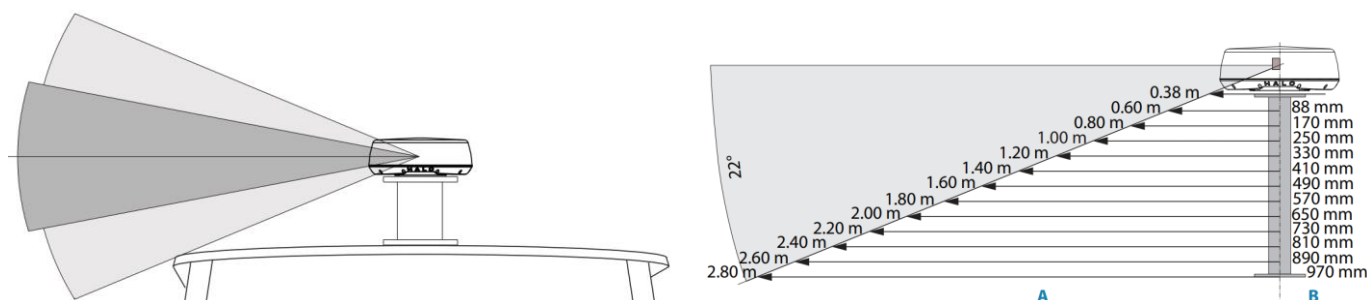
Do not install the scanner:

- Too high up which may cause degradation of the radar picture over short ranges.
- Close to lamps or exhaust outlets. The heat emissions may damage the dome. Soot and smoke will degrade the performance of the radar.
- Close to the antennas of other equipment such as direction finders, VHF antennas, GPS equipment, as it may cause or be subject to interference.
- Where a large obstruction is at the same level as the beam, the obstruction is likely to generate false echoes and/or shadow zones.
- Where it will be subjected to strong vibrations. Vibrations could degrade the performance or service life of the radar.
- Directly on to a large flat roof area. Use a pedestal to elevate the scanner for radar beams to clear roof line.



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For best performance, the radar should be positioned to allow the beams to clear the superstructure of the boat. For every increase of 200 mm (7.87”) of dimension (A), increase the height (B) by 80 mm (1.6”).



• RADAR SCANNER MOUNTING PROCEDURE

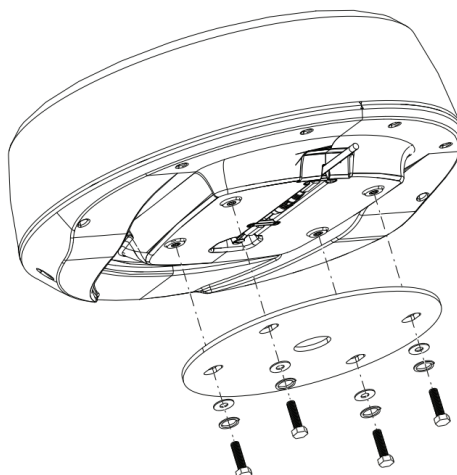
The SIMRAD HALO24 radar scanner is installed on a dedicated steel mast structure at an elevated position above the deck to provide unobstructed 360° radar coverage and maximum detection range. The mast arrangement ensures the scanner is clear of surrounding structures such as handrails, deck equipment, cranes, and accommodation blocks that could otherwise cause shadow sectors or signal reflections.

Prior to installation, the supplied mounting template shall be positioned and secured on the mast top plate, ensuring the scanner front is aligned toward the vessel's bow. Adequate clearance shall be maintained around the scanner to prevent obstruction of the radar beam and to allow safe servicing access.

Use the supplied mounting template and secure it firmly at the selected installation location. Prior to drilling, ensure the following:

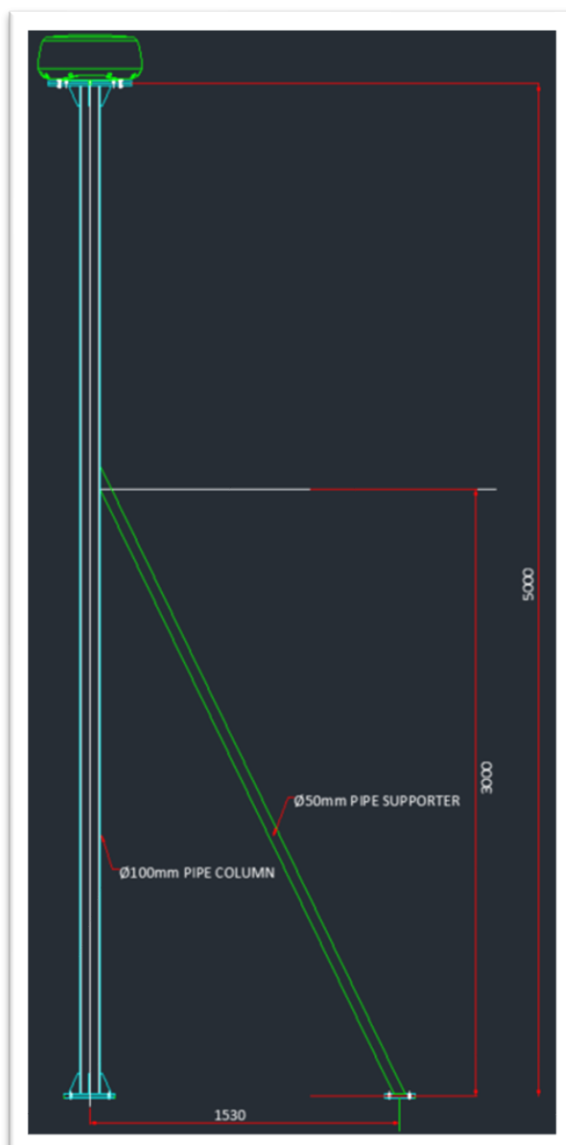
- The template is correctly oriented so that the front of the radar scanner faces the vessel's bow.
- A minimum clearance of 315 mm (12.5 in) is maintained in all directions from the scanner center to avoid obstruction of the radar beam.
- The mounting surface thickness does not exceed 18 mm (0.7 in). If thicker, longer mounting bolts must be used.

Note: Supplied mounting bolts are M8 × 30 mm. If longer bolts are required, use marine-grade stainless steel hardware and ensure a thread engagement between 8 mm and 18 mm.



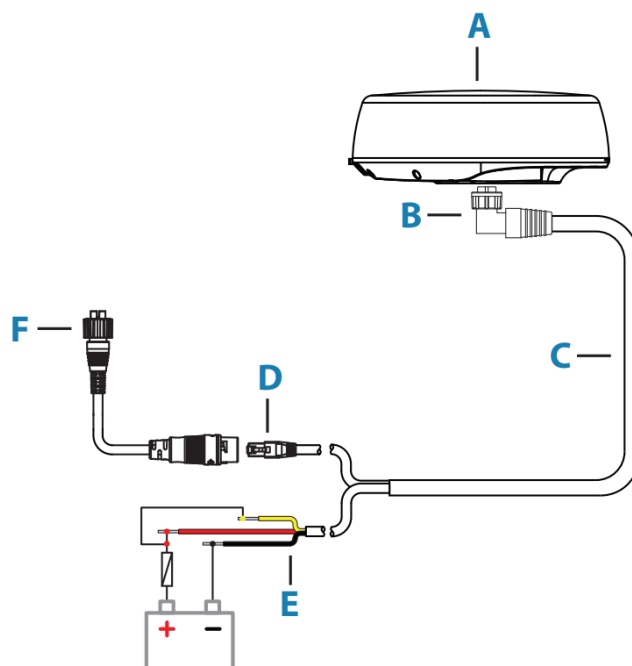
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- Drill four mounting holes using a 9.5 mm (3/8 in) drill bit according to the template.
- Connect the radar scanner interconnection cable before securing the unit.
- Route the cable through the designated cable retention channel to prevent strain or damage.
- Carefully position the scanner over the drilled holes and align with the threaded mounting points.
- Fit a plain washer and lock washer onto each mounting bolt.
- Insert and tighten the bolts securely into the scanner mounting holes.(Tightening torque: 12–18 Nm (8.9–13.3 lb-ft))
- Install triangular support braces on the mast structure to provide adequate rigidity and minimize vibration during vessel operation.
- Route all scanner cables vertically along the mast and secure them at regular intervals using clamps or saddles to prevent mechanical stress.
- Install a junction box below the scanner to provide protected cable termination and allow easy maintenance access.
- Position the installation to minimize shadowing or obstruction from ship structures and maintain safe working clearance for inspection.



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- WIRING INSTRUCTIONS



Key	Description
A	Halo Radar
B	Scanner connector
C	Interconnection cable
D	Ethernet connector
E	Power wires
F	Ethernet adapter RJ45 to 5-pin

Wiring Guidelines	
Do not	
Make sharp bends in the cables	
Run cables in a way that allows water to flow down into the connectors	
Run the data cables adjacent to radar, transmitter, or large/high current carrying cables or high frequency signal cables	
Run cables so they interfere with mechanical systems	
Run cables over sharp edges or burrs	
Attach a mouse line to the Ethernet cable or connector directly as this will loosen and damage the connector and the radar will malfunction when power is applied	
Bend the Ethernet cable back on itself as this will cause damage to the Ethernet cable and the radar will malfunction when power is applied	
Do	
connect a mouse line to the outer jacket of the cable so that the strain of pulling is transferred to the stronger outer jacket of the cable. Use some small cable ties to secure the mouse line to the outer jacket as well if there is sufficient clearance	

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tape the RJ45 connector to the mouse line so that it does not get caught and bent backwards

cover the RJ45 and the conductors with tape to protect them as they are pulled thorough the run

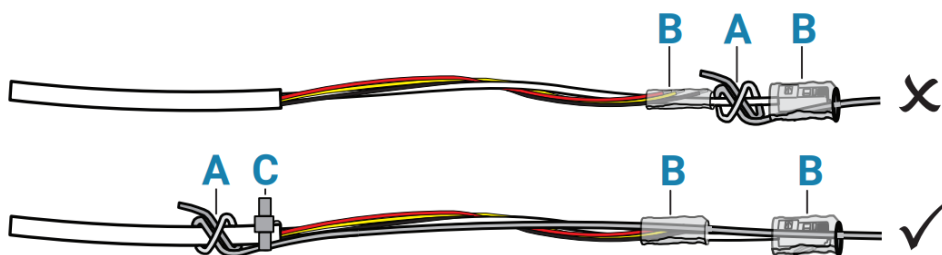
make drip and service loops

use cable ties on all cables to keep them secure

solder/crimp and insulate all wiring connections if extending or shortening the cables

keep joins as high as possible to minimize possibility of water immersion

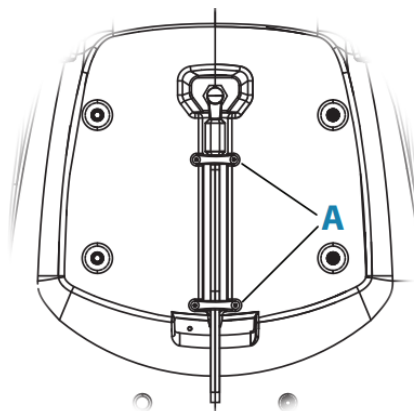
leave room adjacent to connectors to ease plugging and unplugging of cables



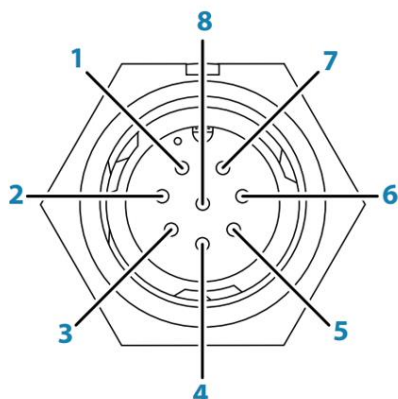
Key	Description
A	Mouse line
B	Electrical tape
C	Cable tie

- SCANNER CONNECTION

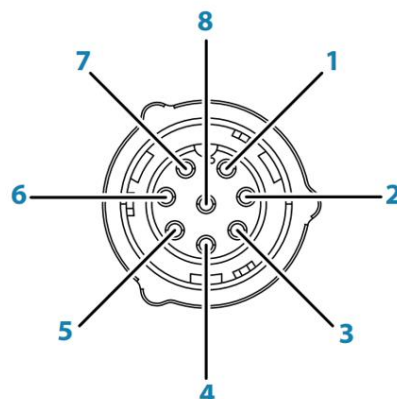
- Insert cable connector on to the male 8 pin plug on the scanner. Take care to align the connector correctly to avoid bending the pins
- Secure the locking collar by rotating clockwise until it clicks
- Feed and secure the cable into the cable retention channel
- Install the 2 cable retainers (A), the screws should be gently tightened



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Scanner socket

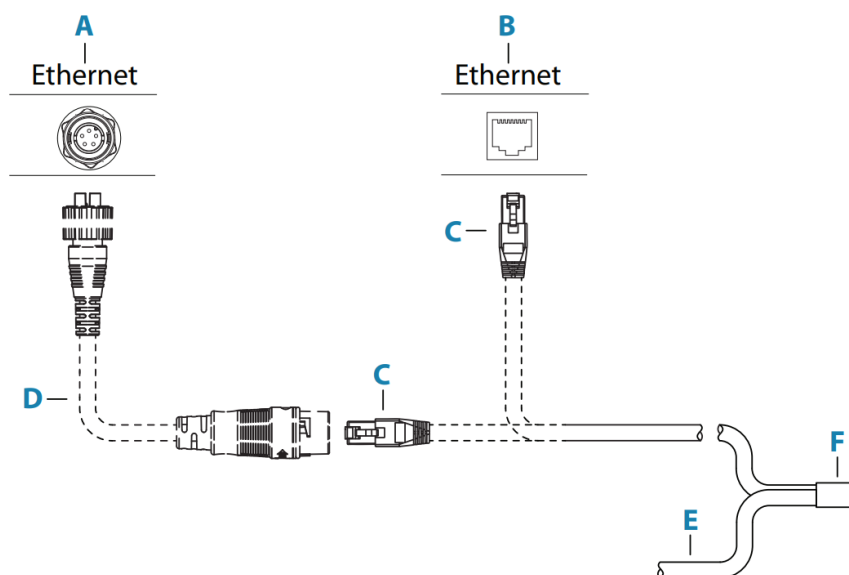


Interconnection cable (scanner end)

Pin-out	Wire color	Description
1	Black	DC negative
2	Yellow	Power control
3	Green	Data Receive -
4	White / Green	Data Receive +
5	Orange	Data Transmit -
6	White / Orange	Data Transmit +
7	Red	+12/24 V DC
8	Drain	Shield

- ETHERNET CONNECTION

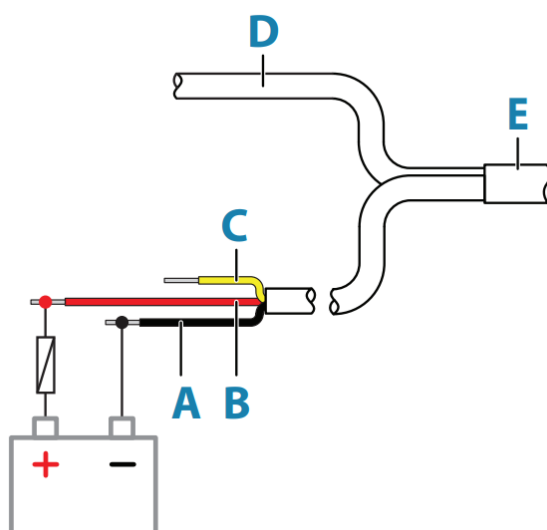
The scanner can be connected to a 5-pin Ethernet connector via the supplied Ethernet adapter.



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Key	Description
A	Display unit or Ethernet switch with a 5-pin Ethernet socket
B	Display unit or Ethernet switch with a RJ45 Ethernet socket
C	Ethernet cable plug (RJ45)
D	Ethernet adapter cable (RJ45 to 5-pin)
E	Power and power control wires
F	Interconnection cable to scanner

- POWER CONNECTION



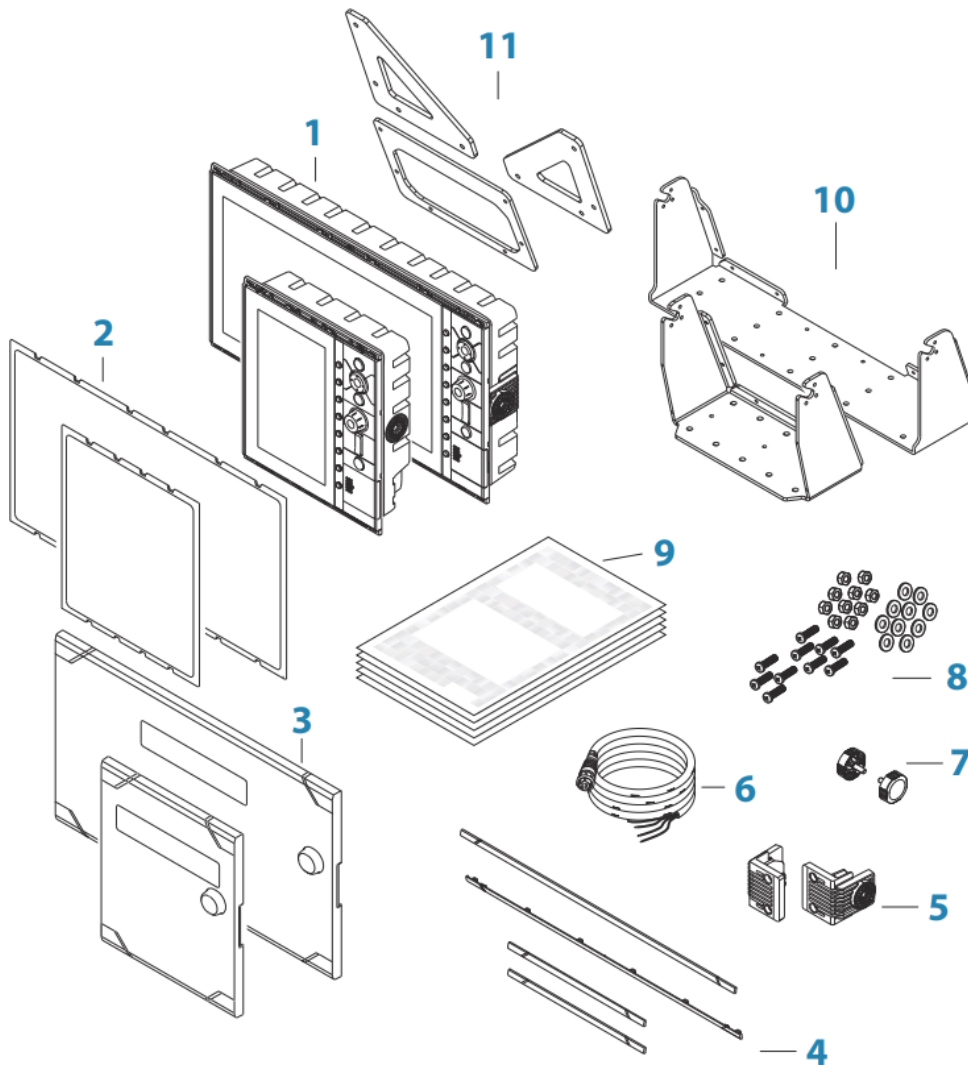
Key	Description
A	DC negative
B	+24 V DC
C	Power control wire
D	Data cable
E	Interconnection cable to scanner

3.1.2 INSTALLATION INSTRUCTIONS OF RADAR CONTROL UNIT WITH DISPLAY SIMRAD -R3016

The SIMRAD R3016 is a dedicated radar control unit and display designed to serve as the primary operator interface for compatible marine radar systems, such as the HALO series. It features an integrated high-definition 16-inch widescreen display that presents a clear and comprehensive radar picture for navigation and situational awareness. The unit incorporates an intuitive keypad and rotary dial interface with direct-access menu keys to provide reliable control of radar functions even in challenging sea conditions. It supports versatile installation options, including flush or bracket mounting, and is engineered for ease of installation and day-to-day operation on board.



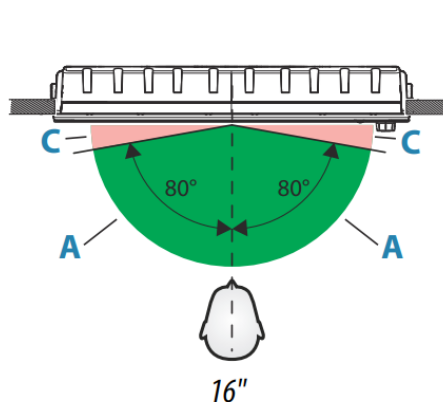
• WHAT'S IN THE BOX



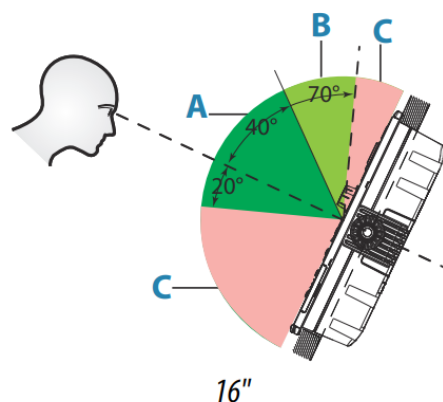
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Key	Description
1	Control unit
2	Panel mounting gasket
3	Sun cover
4	Bezels
5	Gimbal inserts
6	Power cable
7	Knobs
8	Fixing screws
9	Documentation pack
10	U-brackets
11	U-bracket straps

- VIEWING ANGLE



Horizontal viewing angles



Vertical viewing angles

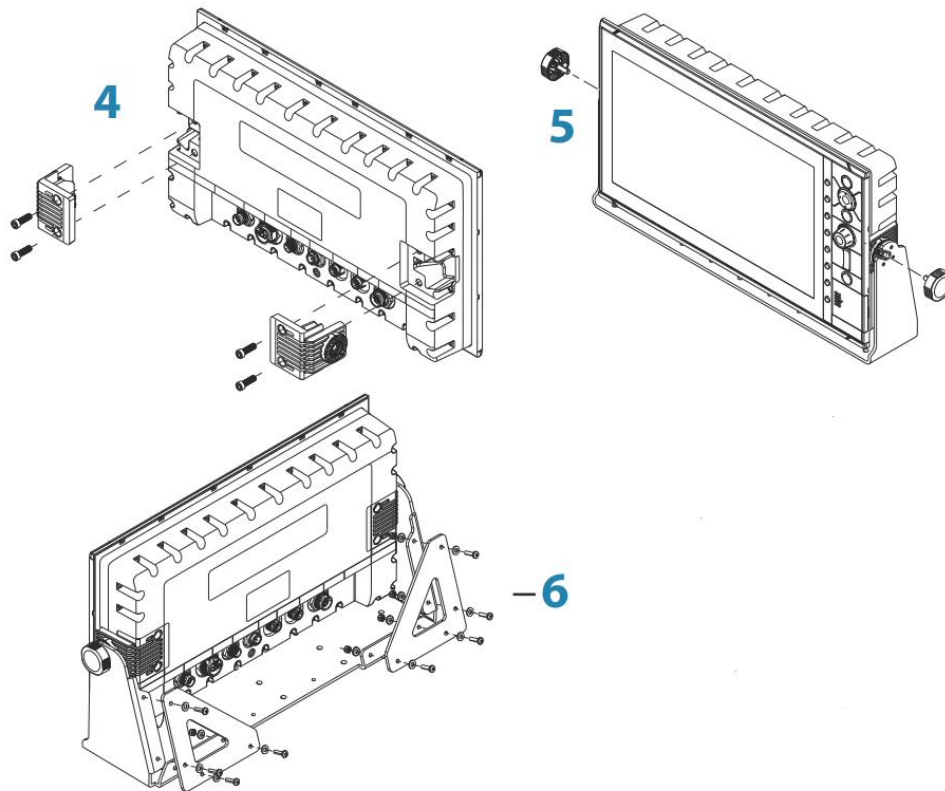
Key	Description
A	Optimum viewing angle
B	Good viewing angle
C	Poor viewing angle or obstructed view

- U-BRACKET MOUNTING

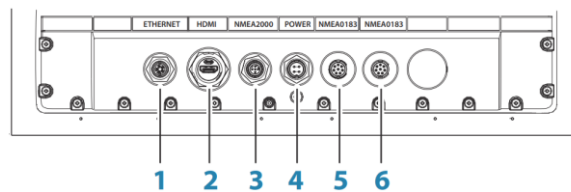
Step	Description
1	Place the bracket in the desired mounting location. Ensure that the chosen location is enough height to accommodate the unit fitted in the bracket and allows tilting of the unit. Also, adequate space is required on both sides to allow tightening and loosening of the knobs
2	Mark the screw locations using the bracket as a template, and drill pilot holes. Use fasteners suited to the mounting surface material. If the material is too thin for self-tappers, reinforce it, or mount the bracket with machine screws and large washers. Use only 304 or 316 stainless steel fasteners.
3	Screw down the bracket.

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4	Using the screws provided in the gimbal kit, fasten the gimbals to the unit.
5	Mount the unit to the bracket using the knobs. Hand tightens only. The ratchet teeth in the bracket and display case ensure a positive grip and prevent the unit changing from the desired angle.
6	Fix the bracket straps.



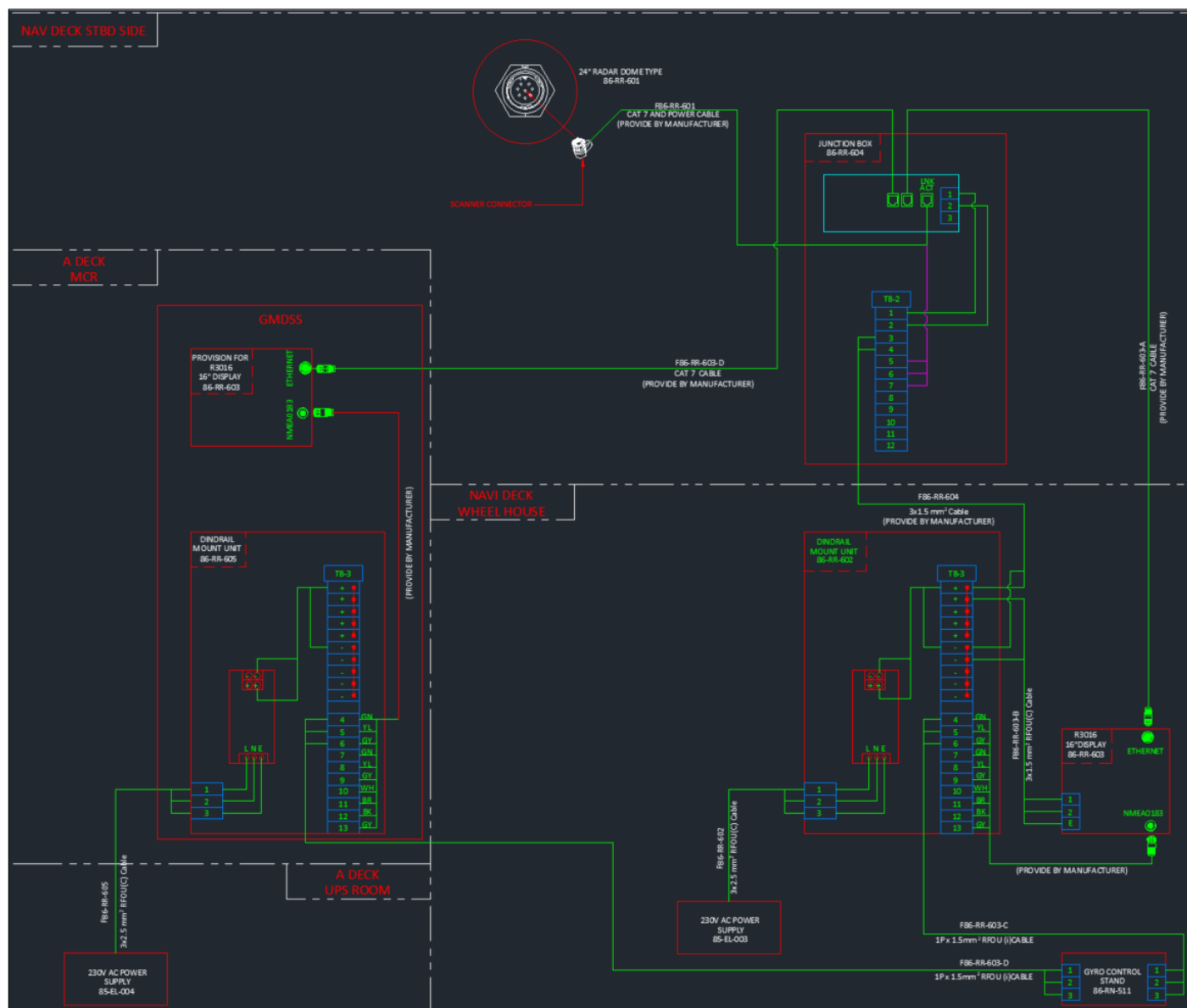
• REAR CONNECTIONS



Key	Description
1	Ethernet, 5-pin
2	HDMI
3	NMEA 2000, 5-pin
4	Power, 4-pin
5	NMEA 0183 (Serial 1)
6	NMEA 0183 (Serial 2)

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3.2 OVERALL SYSTEM ARCHITECTURE



The radar system is arranged as an integrated network comprising a mast-mounted scanner, deck-level junction and termination facilities, centralized power distribution, and an indoor control and display position located within the navigation spaces. The architecture is designed to provide reliable radar operation, simplified cable routing, and flexibility for both commissioning and final operational configuration.

The radar scanner is installed externally at the navigation deck starboard side mast head to achieve maximum radar horizon and unobstructed 360° coverage. From the scanner, a combined power and Ethernet communication cable is routed vertically along the mast and secured at regular intervals to prevent mechanical stress and vibration. The cable is led into a nearby weather-protected junction box where termination and distribution of power and data connections are performed.

From the junction box, the Ethernet communication line and DC power supply are extended through marine-grade cables to the indoor equipment locations. The Ethernet link carries both radar video data and control signals between

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the scanner and the radar control/display unit, while the dedicated DC supply provides stable operating power to the radar equipment.

Primary AC power is sourced from the vessel's 230 VAC UPS distribution board. This supply is routed through protective breakers and converted to the required DC voltage before being delivered to the radar system, ensuring continuous and protected operation during both normal and backup power conditions.

The radar control and display position is presently installed at the Navigation Deck Wheelhouse to support commissioning, testing, and navigation during construction and transit. A permanent provision for the control unit is included within the GMDSS console. Upon FPSO anchoring and commencement of operational service, the control unit will be relocated from the wheelhouse to the GMDSS panel, utilizing the pre-installed cabling and interfaces without modification to the external scanner or junction arrangements.

Navigation sensor inputs, including heading and other interfaced signals, are routed to the control position to enable stabilized and accurate radar presentation. All cabling is arranged through dedicated trays and conduits to maintain segregation between power and signal circuits, reduce interference, and allow safe maintenance access.

This distributed architecture ensures:

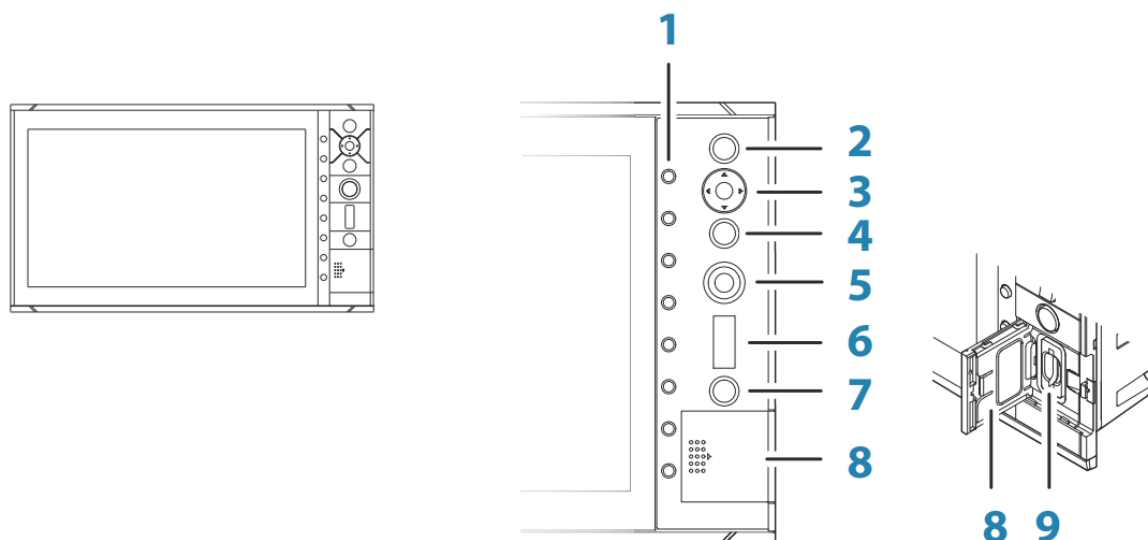
- Clear radar coverage from an elevated scanner position
- Short and protected external cable runs
- Centralized indoor control and monitoring
- Reliable power from the UPS supply
- Flexible relocation of the display/control unit
- Simplified maintenance through junction and termination points

The overall design therefore provides a robust, serviceable, and operationally adaptable radar installation suitable for both temporary navigation use and permanent FPSO operational deployment

3.3 OPERATION

3.3.1 OPERATION INSTRUCTIONS OF R3016 RADAR CONTROL UNIT

- FRONT PANEL AND KEYS



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Key	Description
1	Softkeys
2	Enter (ENT) key
3	Arrow keys
4	Exit (EXIT) key
5	Rotary knob-Rotate to scroll through menu items and to adjust values, press to select or to save settings.
6	Range (RANGE) key- Press the + or the - indication to increase or decrease the radar range.
7	Standby/Brilliance (STANDBY/BRILL) key- Press once to display the Brilliance/Standby pop-up, press again to toggle between Standby and Transmit mode. Press and hold to switch the radar system on/off.
8	Card reader door
9	SD card reader

- MAIN PANEL



Key	Description
1	Plan Position Indicator (PPI) - Radar video area where all tracking and navigation options are performed.
2	Own ship information -Stabilization mode indicator, picture freeze indicator and gauges showing primary and secondary sensors.
3	Target panel -Detailed information about selected targets and AIS targets.
4	Softkey bar -Reference for softkey functions.
5	Target indicators -Overview of target indicator settings.
6	Markers -Details for active VRM and EBL markers.

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7	Cursor information -Range and bearing from the vessel to the cursor position. Also include position information if a position source is available.
8	Alerts panel -List of all active alerts.
9	Signal indicators -Gauges for signal processing and indicators for radar functions.
10	System information -Range, mode and pulse details

Power ON / OFF

- Press the STANDBY/BRILL key to switch the system ON.
- Press and hold the STANDBY/BRILL key for 5 seconds to switch OFF both the control unit and radar antenna.
- If configured as a power slave, the unit turns ON/OFF automatically with the power master.

Display Brightness (Brilliance) Adjustment

- Press the STANDBY/BRILL key to open the Brilliance pop-up.
- Rotate the rotary knob to increase or decrease brightness.
- Brightness setting is automatically saved for next startup.
- System provides Day and Night colour palettes: Above 40% → Day mode/40% or below → Night mode
- Press EXIT to close the pop-up.

Radar Source Selection

- Multiple radar antennas can be connected.
- Select the active antenna/source from the main menu.

Transmit / Standby Mode

- Open the Brilliance/Standby pop-up.
- Press STANDBY/BRILL key to toggle between:
 - Transmit (TX) – radar actively scanning
 - Standby – radar transmission stopped

Rader Range Adjustment

- On the Range key Press Range (+) to increase range (–) to decrease range.
- Current range scale is shown on the upper left corner of the display.
- Available ranges depend on the connected radar antenna.

Cursor Operation

- Press any Arrow key to activate the cursor.
- Use Arrow keys to move the cursor within the radar screen (PPI).
- Press EXIT to deactivate the cursor.
- When active, the display shows: Range and bearing to cursor/ Geographic position

Speed Source Selection

- Speed data can be received from: Speed Log (water/sea speed)-EPFS/GPS (ground speed)
- Select preferred source from Speed Source menu.
- Radar automatically adjusts stabilization mode based on selected source.

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Stabilization Modes

Stabilization mode compensates for vessel movement to maintain a steady and accurate radar picture. The system uses speed information from connected sensors to stabilize the radar display either relative to the sea surface or the ground.

Sea Stabilization

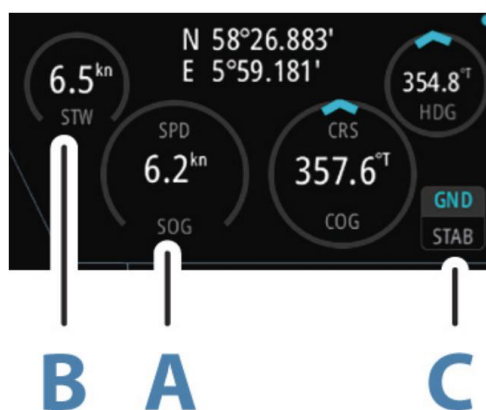
- Uses speed through water from the Speed Log.
- Radar picture is referenced to the water surface.
- Suitable for manoeuvring and short-range navigation.
- May be affected by tide and current, reducing ground position accuracy.

Ground Stabilization

- Uses speed over ground from EPFS/GPS.
- Radar picture is referenced to the earth/ground.
- Fixed objects remain stationary on the display.
- Provides more accurate target tracking and collision avoidance.
- Recommended for normal navigation and offshore operation.

Operation

- Select the preferred speed source from the Speed Source menu.
- The radar automatically applies the corresponding stabilization mode.
- Active mode and source are shown on the display.



Primary speed source (A), secondary speed source (B) and stabilization mode (C) are shown in the Own ship information panel.

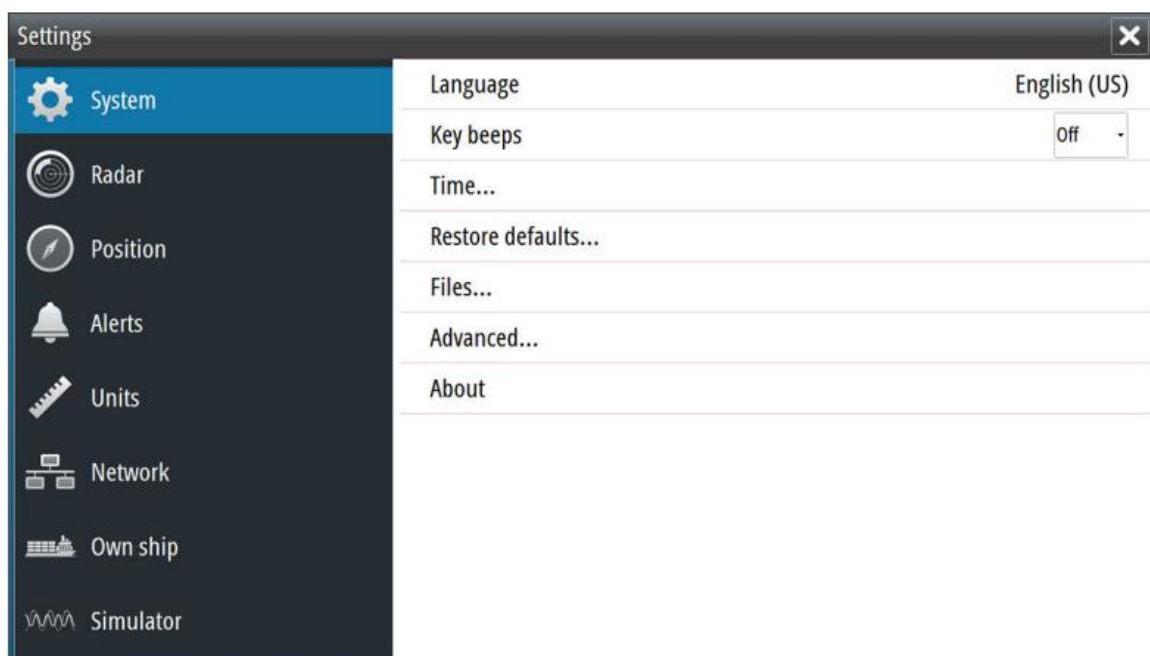
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- Radar Software Setup

Press the Menu button on the radar control unit. Select Settings. Below are the main areas to configure before first use.

- System Settings
- Network Settings
- Radar Settings
- Own Ship Settings

System Settings

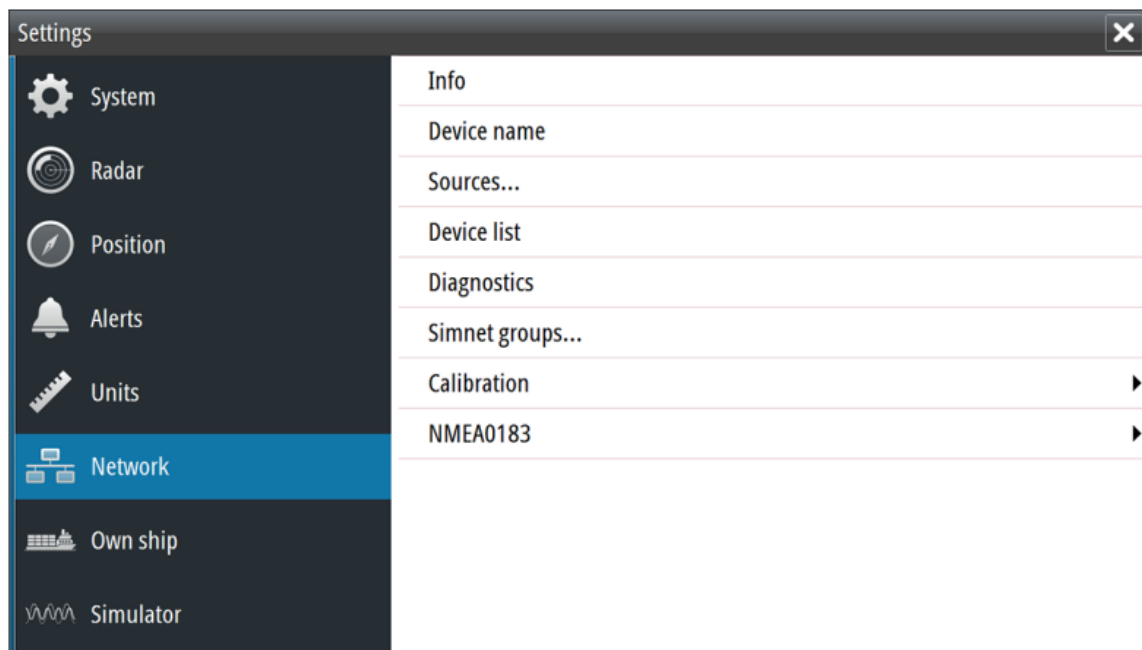


- Language: Choose your preferred language
- Key Beeps: Adjust volume of button presses.
- Time: Set correct time zone and time format.
- Restore Defaults: Only use if you want to reset to factory settings.
- Files: Browse internal memory and SD card. Useful for updates.
- Advanced: Enable or hide features in menus.
- About: Check software version and copyright info.

Purpose of the system setting is to set basic system behavior and preferences. Make sure system restarts after changing language or time.

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Network Settings



- Device Name: Give a unique name if multiple units are installed.
- Data Sources: Select internal (GPS, sonar) or external (NMEA 2000/0183) sources.
- Auto Select: Automatically picks the highest priority source.
- Manual Selection: Use only if multiple sources exist and auto selection is incorrect.
- Device List: Shows all connected devices; assign instance numbers for identical devices.
- Diagnostics:
 - Check Rx/Tx errors, bus load, and device communication.
 - Correct physical network issues if errors are high (cable lengths, terminations, number of nodes).
- SimNet Groups: If multiple units are installed, group them to apply settings to all simultaneously.
- Calibration: Apply offsets to speed, temperature, depth, or pressure from sensors.
- NMEA 0183: Set baud rate and select which sentences to transmit/receive.

Radar Settings

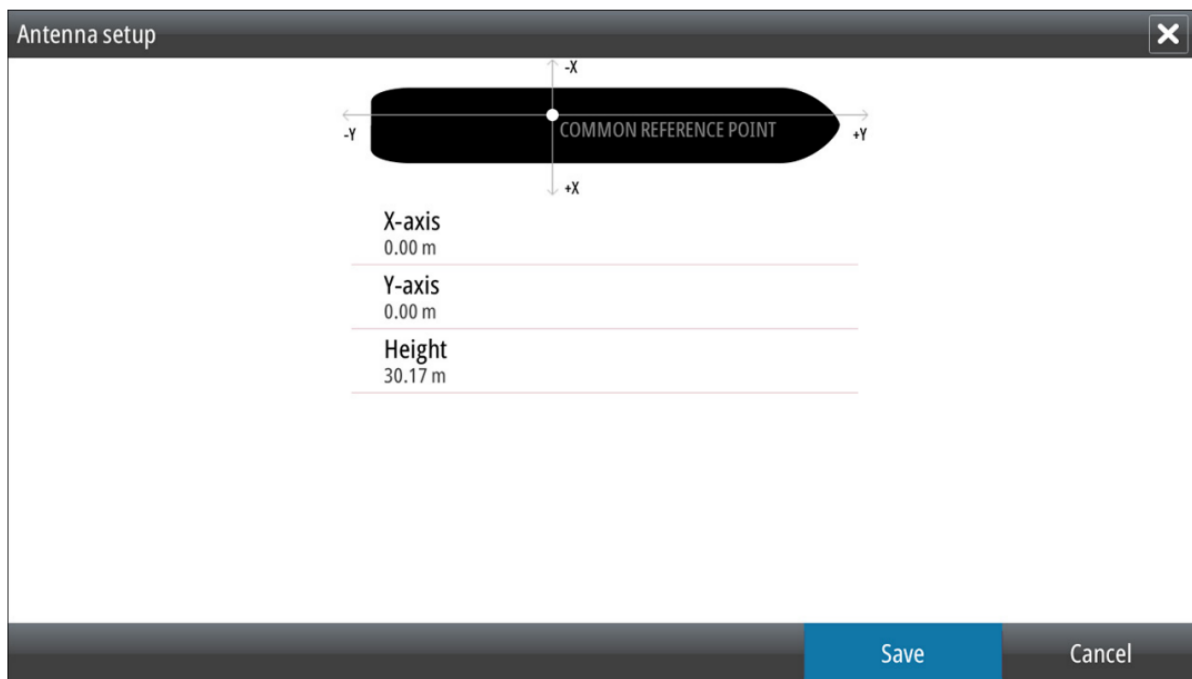


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Radar Source: Select which antenna you want to configure.

Antenna Setup:

- Enter position and height of antenna on the vessel.
- Important for accurate sea clutter handling and target plotting.



Antenna setup

Diagram: A vessel hull is shown with a central point labeled "COMMON REFERENCE POINT". Axes are indicated: -X (up), +X (down), -Y (left), and +Y (right).

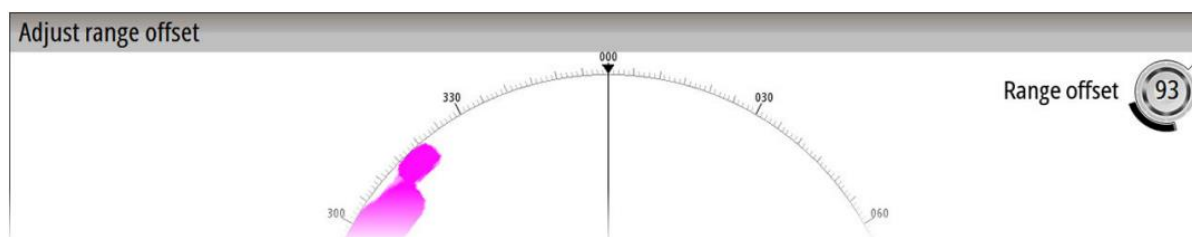
Input fields:

- X-axis: 0.00 m
- Y-axis: 0.00 m
- Height: 30.17 m

Buttons: Save, Cancel

Range Offset:

- Turn the rotary knob to adjust the start of radar sweep.
- Stand ~50–100 yards from a straight jetty → adjust range so jetty appears straight.



Bearing Alignment:

- Point ship at a stationary object.
- Rotate heading line until it aligns with object.

Sector Blanking:

- Block unwanted radar reflections from masts or nearby structures.
- Use the sector blanking feature to stop the radar from transmitting in the direction up to four sectors

Sidelobe Suppression:

- Reduce false echoes near strong targets.
- Set range to 0.5–1 NM, start with Auto, adjust manually until false echoes disappear.

Tuning:

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- Adjust for maximum target clarity using large, identifiable objects.

Local Interference Rejection:

- Filter out local radar noise.

Open Array Park Angle:

- Set resting antenna position when radar is in standby.

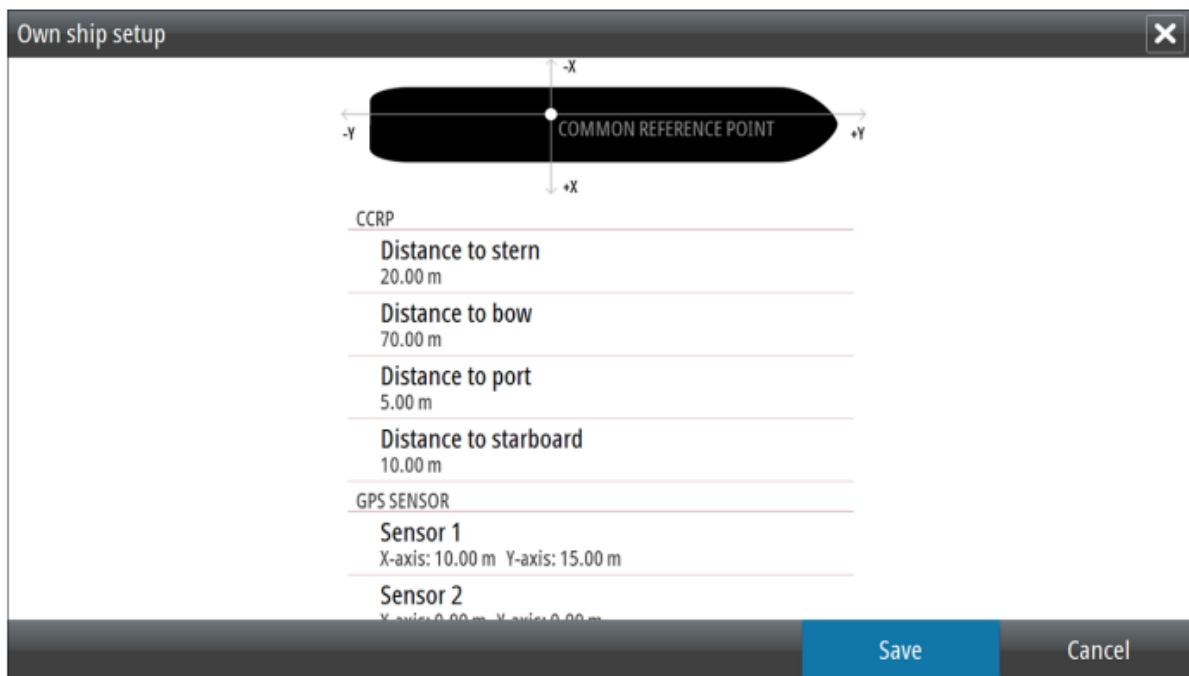
Reset Radar:

- Restore factory defaults if needed.

Radar Palettes:

- Choose colour for day use; night colour adjusts automatically.

Own Ship Setup



Purpose: Ensure radar knows where your vessel and GPS are located.

- CCRP (Common Reference Point):
 - Usually set at bridge/helm where radar operator is.
 - Radar calculates bearing & distance of targets relative to this point.
- GPS Sensor Location:
 - Set primary GPS sensor position.
 - Ensures AIS and other targets appear in correct position.
- MMSI:
 - Enter vessel MMSI number.
 - Prevents your own vessel from showing as an AIS target.

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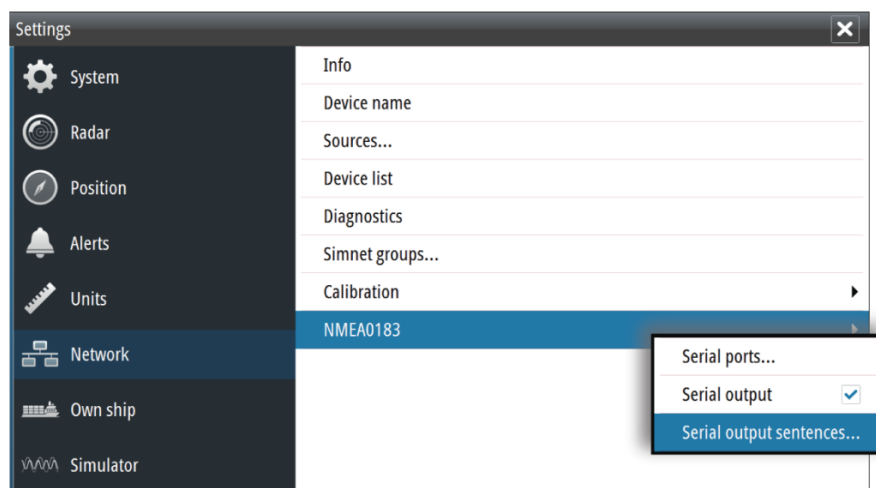
3.4 MAINTENANCE

3.4.1 MAINTENANCE OF SIMRAD- HALO24

Clean the radome using soapy water and a soft cloth. Avoid using abrasive cleaning products. Do not use solvents such as gasoline, acetone, M.E.K etc. as this will damage the dome surface.

3.4.2 MAINTENANCE OF R3016 RADAR CONTROL UNIT

- Preventive Maintenance
 - The unit contains no user-serviceable internal parts. Only basic external inspection and cleaning are required.
- General Care
 - Fit the protective sun cover when the unit is not in use.
 - Protect the unit from prolonged UV exposure and harsh weather.
- Display Cleaning
 - Clean screen with a soft cloth and fresh water to remove salt.
 - For stains, use 50/50 warm water and isopropyl alcohol.
 - Do not use solvents, ammonia, or abrasive cleaners.
 - Apply minimal pressure to avoid scratching.
- Media Port Door
 - Clean regularly to prevent salt buildup and water ingress.
- Keys and Controls
 - Check that keys move freely and are not stuck.
 - Free any jammed keys gently.
- Connectors
 - Inspect visually for damage or looseness.
 - Ensure plugs are fully inserted and locked.
- Data Logging
 - System automatically records NMEA data logs for troubleshooting.
 - Logs can be exported to external storage when required.



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- Software and Backup
 - Update software using the latest files from the manufacturer.
 - Regularly back up system settings to external media.

3.5 TROUBLESHOOTING GUIDE

The radar control unit continuously monitors the system for faults, danger situations, and operational alerts while running. When an alert condition occurs, it is displayed in the Alerts panel along with a descriptive text and appropriate icon. Alerts remain visible until the underlying condition is resolved and acknowledged by the operator.

There are three main alert types with differing priorities and indications:

- **Alarms:** Highest priority conditions that may affect safety or system performance.
 - Accompanied by an audible signal.
 - The alert icon flashes until acknowledged.
- **Warnings:** Medium priority conditions indicating issues that may lead to problems if unaddressed.
 - Accompanied by periodic audible signals.
 - The warning icon flashes until acknowledged.
- **Cautions:** Informational notifications that require attention but do not trigger audible alerts.
 - Display only a steady icon and descriptive text

When an alert is triggered, the alert appears in the Alerts panel. Alarms and warnings remain in the Alerts panel until the reason for the alarm/warning is removed and the alarm/warning has been acknowledged.

Alert type	Icon	State	Indication
Alarm		Active - not acknowledged, not silenced	• Flashing symbol and descriptive text • Audible signal
		Active - acknowledged	• Steady symbol and descriptive text • No audible signal
		Rectified - not acknowledged	• Flashing symbol and descriptive text • No audible signal

Alert type	Icon	State	Indication
Warning		Active - not acknowledged, not silenced	- Flashing symbol and descriptive text - Audible signal
		Active - not acknowledged, silenced	- Flashing symbol and descriptive text - No audible signal
		Active - acknowledged	- Steady symbol and descriptive text - No audible signal
		Active - responsibility transferred	- Steady symbol and descriptive text - No audible signal
		Rectified - not acknowledged	- Flashing symbol and descriptive text - No audible signal
Caution		Active	- Steady symbol and descriptive text - No audible signal

Use the Alerts softkey (labelled “Alerts Acknowledge” or “Alerts Ack”) to acknowledge notifications. Pressing the soft key moves the cursor to the Alerts panel and allows navigation through active alerts. Repeated presses acknowledge alerts from newest to oldest. Press EXIT to exit the Alerts panel. When an alarm or warning is acknowledged, the audible signal stops if there are no remaining unacknowledged alerts. The Alerts dialog includes both active alerts and a history of recent alert state changes. Each alert entry is time-stamped, assisting in diagnostics and troubleshooting

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Action	Symptom	Remedy
Unknown Radar	Generally seen when overlaying the radar on the chart.	Ensure that the correct radar source is selected and configured in the display unit.
No radar	Indicates the display does not see the radar.	- Check/select radar in Radar Sources - Power cycle the system - Check all connections - Check voltage to the yellow power control wire - Insufficient power supply - Missing or intermittent Ethernet connection, check for damage of the RJ45 connector - Faulty Ethernet cable - Try another Ethernet port on the display or switch
No scanner	Sometimes seen before the “No Radar” message above:	- Check voltage/current - If persistent, power cycle the system, check the scanner cable / RJ45 - Possible internal fault with the radar, contact service
No spoke data	Sometimes seen before the “No Scanner” message above	- Check voltage/current - If persistent, power cycle the system, check the scanner cable / RJ45 - Possible internal fault with the radar, contact service
Error codes		
Error code	Description	Recommendation
0x00000001	Radar saved settings corrupted	Radar will revert to factory defaults. Re-enter your settings including installation settings
0x0001000C	Scanner not detected	- Check the pedestal interconnection cable connections - Power cycle the radar - Check input Voltage
0x0001000D	Transmitter overheat (soft)	- Try changing to shorter ranges - Switch to STBY, Allow unit cool
0x0001000E	Transmitter overheat (hard)	Switch to STBY, Isolate power to the radar and contact service
0x0001000F	Signal processing error	Unit should revert to STBY. Select transmit If problem persists. power cycle the radar
0x00010017	Scanner failure	Contact service
0x00010010	Power supply overheating	Switch to STBY, allow unit cool then retry
0x00010011	Power supply voltage error	Check scanner cable for connections for corrosion or damage
0x00010012	Power supply overload	Contact service
0x00010013	Power supply hardware fault	Contact service
0x00010014	Power supply comms fault	Contact service

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0x00010019	Low battery voltage (Supply voltage low)	- Recharge and check supply voltage - Restart the rada
0x00010016	LED Lighting fault	Turn accent lighting off and then retry
0x00010018	Radar interface box fault	Check the interconnection cable for damage

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4
PROJECT SPECIFIC DRAWINGS

TELECOM CABLE BLOCK DIAGRAM - NEW RADAR SYSTEM

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
1	15.09.2025	RE-ISSUED FOR CONSTRUCTION	CP	HS	TN	
0	03.09.2025	ISSUED FOR CONSTRUCTION	CP	HS	TN	
A	25.08.2025	ISSUED FOR REVIEW	CP	HS	TN	

COMPANY:



ORIENTAL
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

VENDOR:



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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM CABLE BLOCK DIAGRAM – NEW RADAR SYSTEM

SCALE:	DRAWING NO. 2024-VIVO-EMEM-TS-BD-040	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)BF-029	REV.
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3								3
4								4
5								5
6								6
7								7
A	B	C	D	E	F	G	H	I

BILL OF MATERIAL			
S/N	SYMBOL	DESCRIPTION	QTY
1		24" RADAR DOME TYPE MODEL : HALO 24 MANUFACTURER : SIMRAD	01
2		RADAR DISPLAY UNIT WITH CONNECTION BOX MODEL : 000-12188-001 MANUFACTURER : SIMRAD	01
3		JUNCTION BOX 1 MODEL : OEM MANUFACTURER : VIVO	01

REVISIONS				
1	15.09.2025	RE-ISSUED FOR CONSTRUCTION	CP	HS TN
0	03.09.2025	ISSUED FOR CONSTRUCTION	CP	HS TN
A	25.08.2025	ISSUED FOR REVIEW	CP	HS TN
REV.	DATE	DESCRIPTION	DRWN	CHKD APPD CLIENT

COMPANY:



ORIENTAL
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

VENDOR:



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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

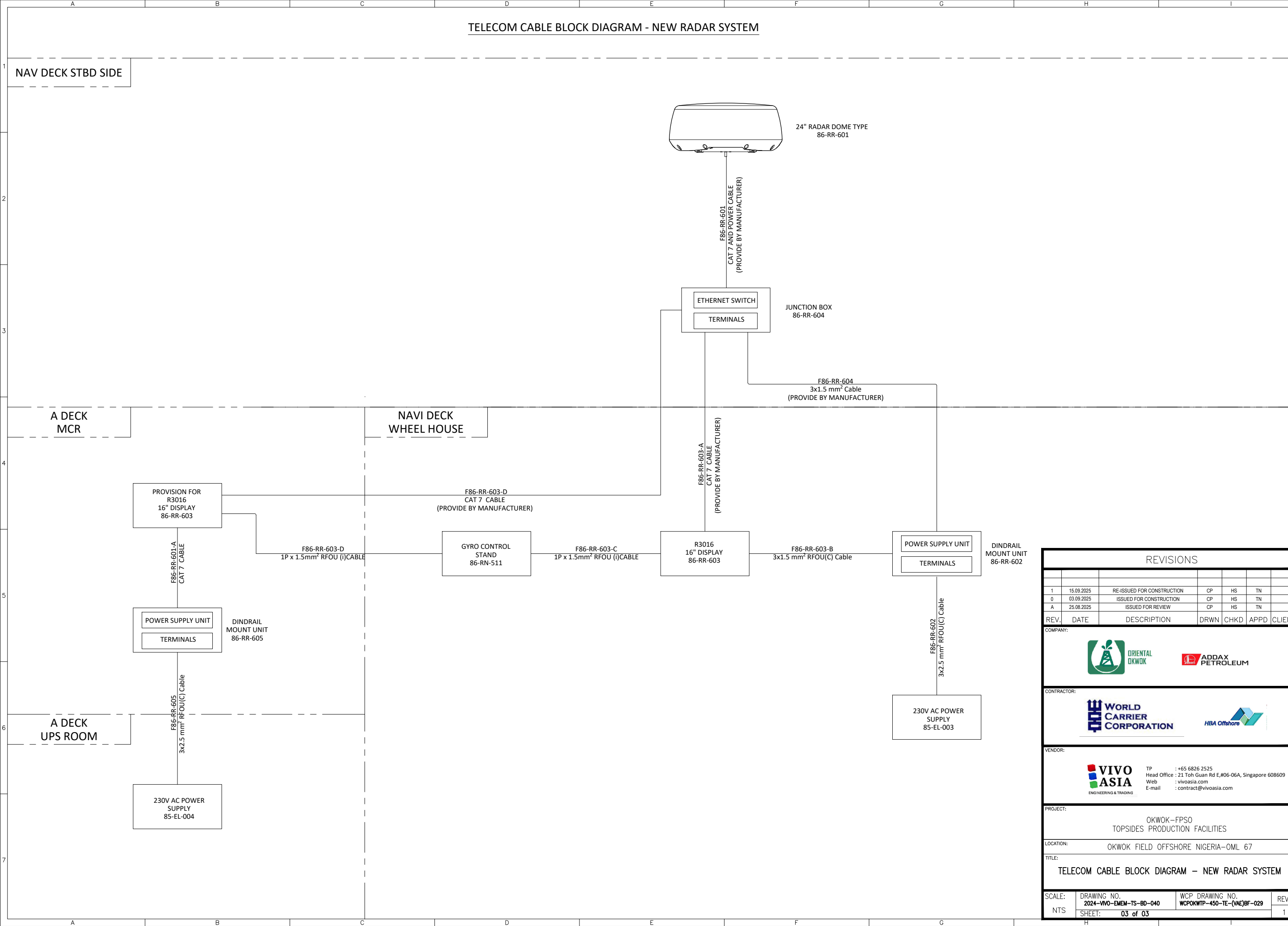
LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM CABLE BLOCK DIAGRAM – NEW RADAR SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
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REVISIONS						
1	15.09.2025	RE-ISSUED FOR CONSTRUCTION	CP	HS	TN	
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 ORIENTAL OKWOK  ADDAX PETROLEUM						
CONTRACTOR:						
 WORLD CARRIER CORPORATION  HBA Offshore						
VENDOR:						
 VIVO ASIA ENGINEERING & TRADING TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com						
PROJECT:						
OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES						
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OKWOK FIELD OFFSHORE NIGERIA-OML 67						
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TELECOM TYPICAL INSTALLATION DRAWING - NEW RADAR SYSTEM

REVISIONS

1	16.09.2025	RE-ISSUED FOR CONSTRUCTION	DR	HS	TN	
0	03.09.2025	ISSUED FOR CONSTRUCTION	CP	HS	TN	
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LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING
-NEW RADAR SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
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2	03.09.2025	16.09.2025		INDEX PAGE	
3	03.09.2025	16.09.2025		RADAR - DIMENSION & MOUNTING DRAWINGS	
4	03.09.2025	16.09.2025		RADAR - DIMENSION & MOUNTING DRAWINGS	
5	03.09.2025	16.09.2025		RADAR - DIMENSION & MOUNTING DRAWINGS	

REFERENCE			
NO	REVISION	CLIENT DOC NO.	TITLE
1	0	WCPOKWTP-450-TE-(VAE)LY-018	Telecom Device Location Layout Drawing- Radar System

REVISIONS

1	16.09.2025	RE-ISSUED FOR CONSTRUCTION	DR	HS	TN	
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PROJECT:

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LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

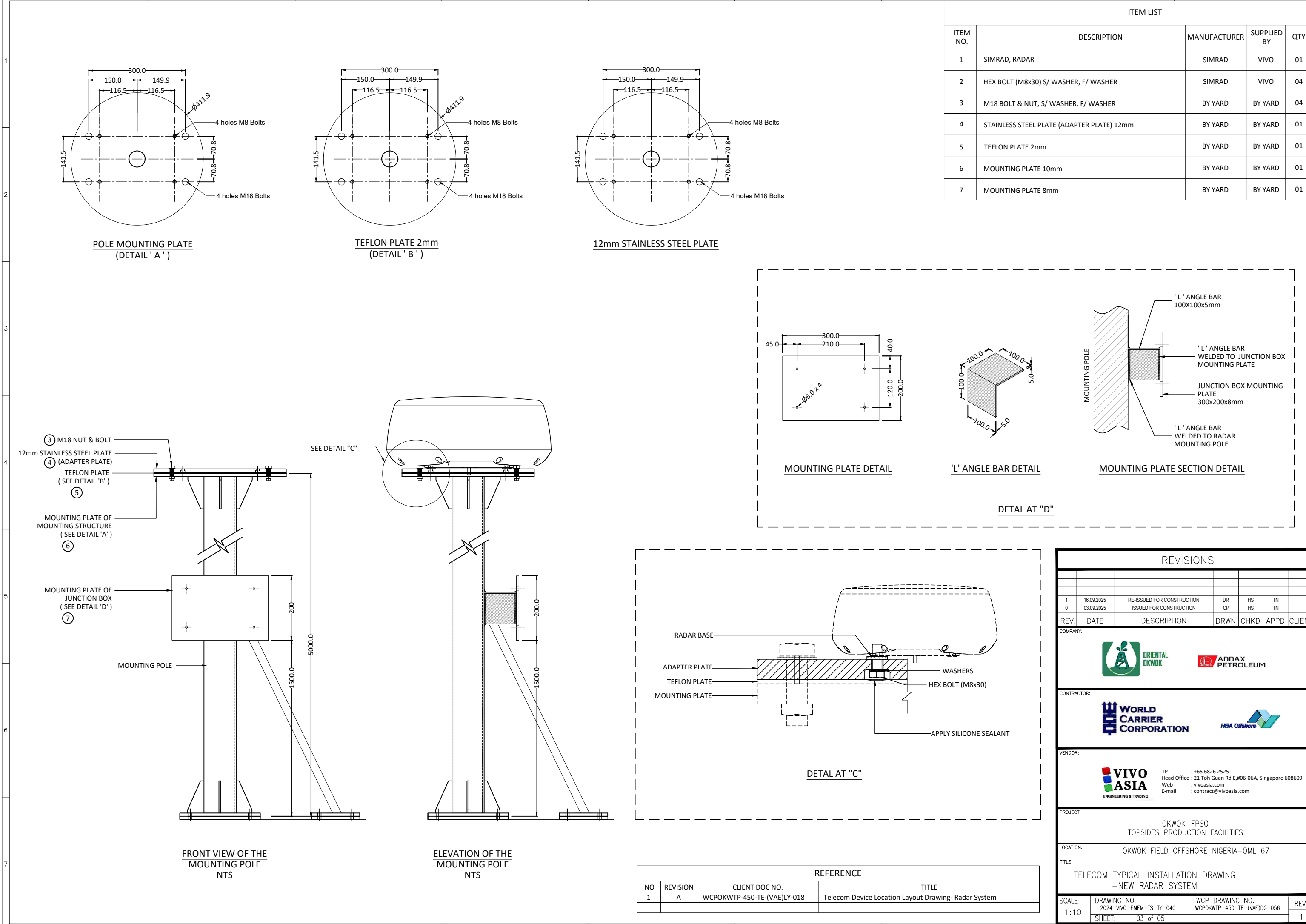
TELECOM TYPICAL INSTALLATION DRAWING
-NEW RADAR SYSTEM

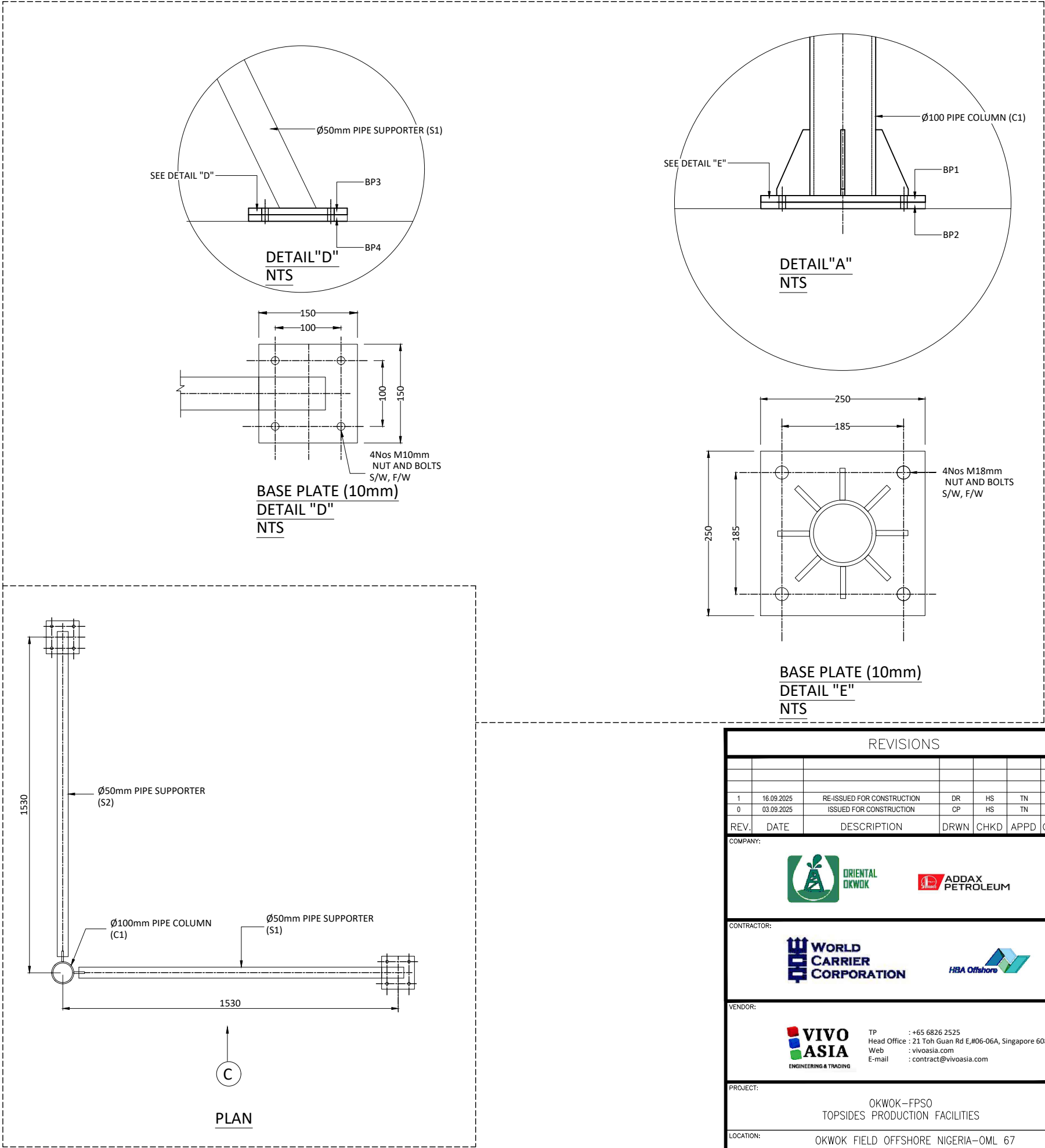
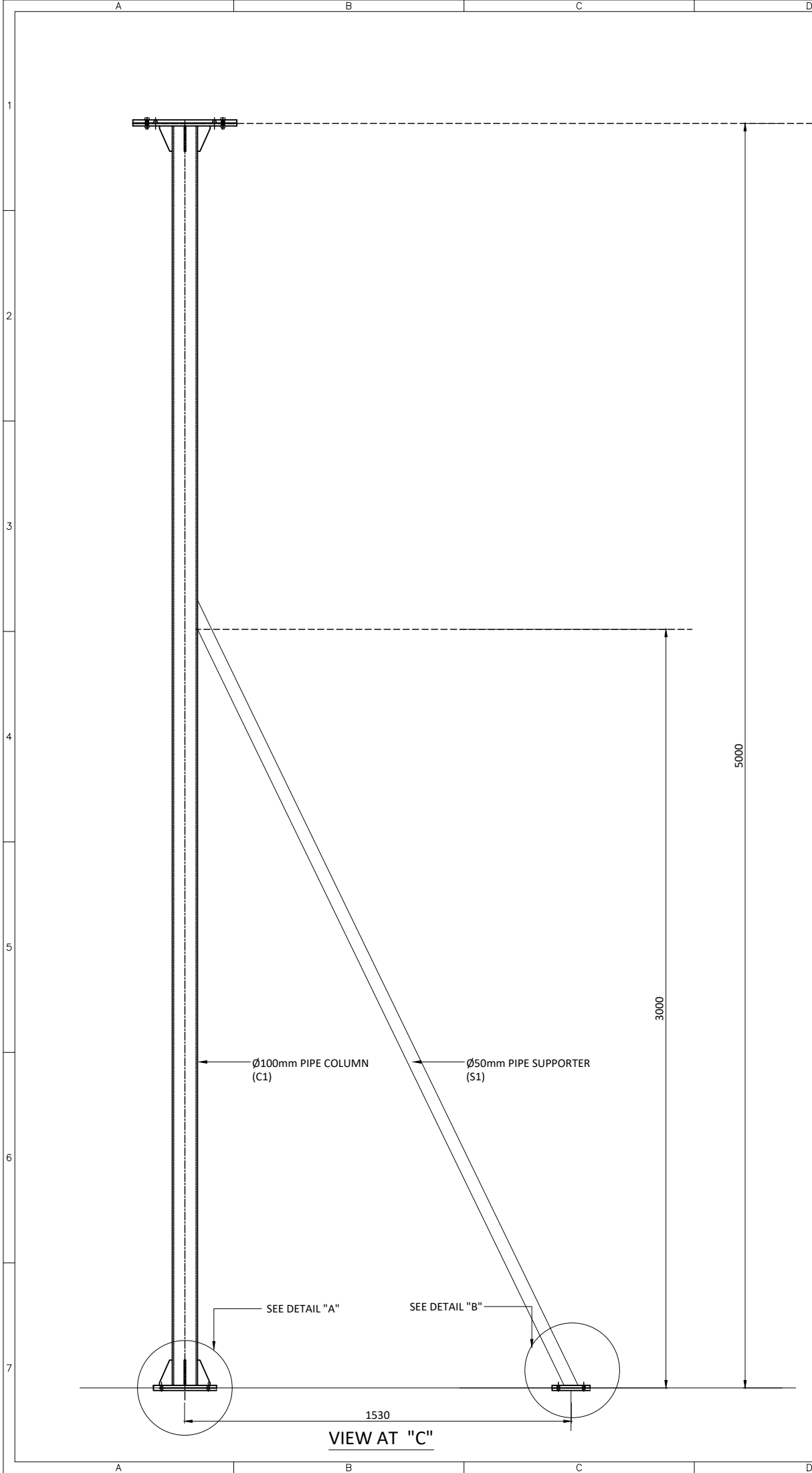
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2024-VIVO-EMEM-TS-TY-040
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WCP DRAWING NO.
WCPOKWTP-450-TE-(VAE)DG-056

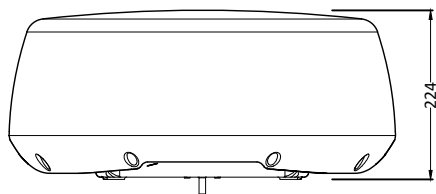
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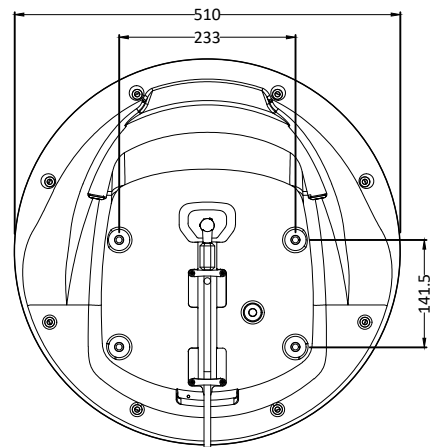


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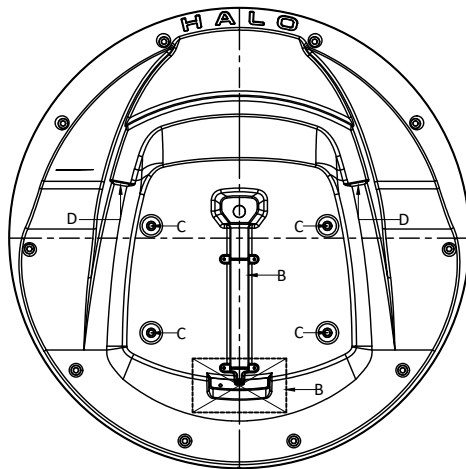
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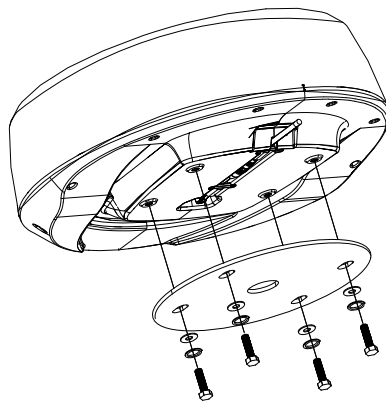
FRONT VIEW



REAR VIEW



REAR VIEW



KEY	DESCRIPTION
A	CABLE ENTRY AREA
B	LED ACCENT LIGHT
C	BOLT HOLE M8X30mm
D	LED ACCENT LIGHT

REFERENCE			
NO	REVISION	CLIENT DOC NO.	TITLE
1	A	WCPOKWTP-450-TE-(VAE)LY-018	Telecom Device Location Layout Drawing- Radar System

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TELECOM TYPICAL INSTALLATION DRAWING
-NEW RADAR SYSTEM

SCALE:

1:10

DRAWING NO.

2024-VIVO-EMEM-TS-TY-040

WCP DRAWING NO.

WCPOKWTP-450-TE-(VAE)DG-056

REV.

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TELECOM TERMINATION DRAWING - NEW RADAR SYSTEM

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	22.09.2025	ISSUED FOR CONSTRUCTION	CP	HS	TN	

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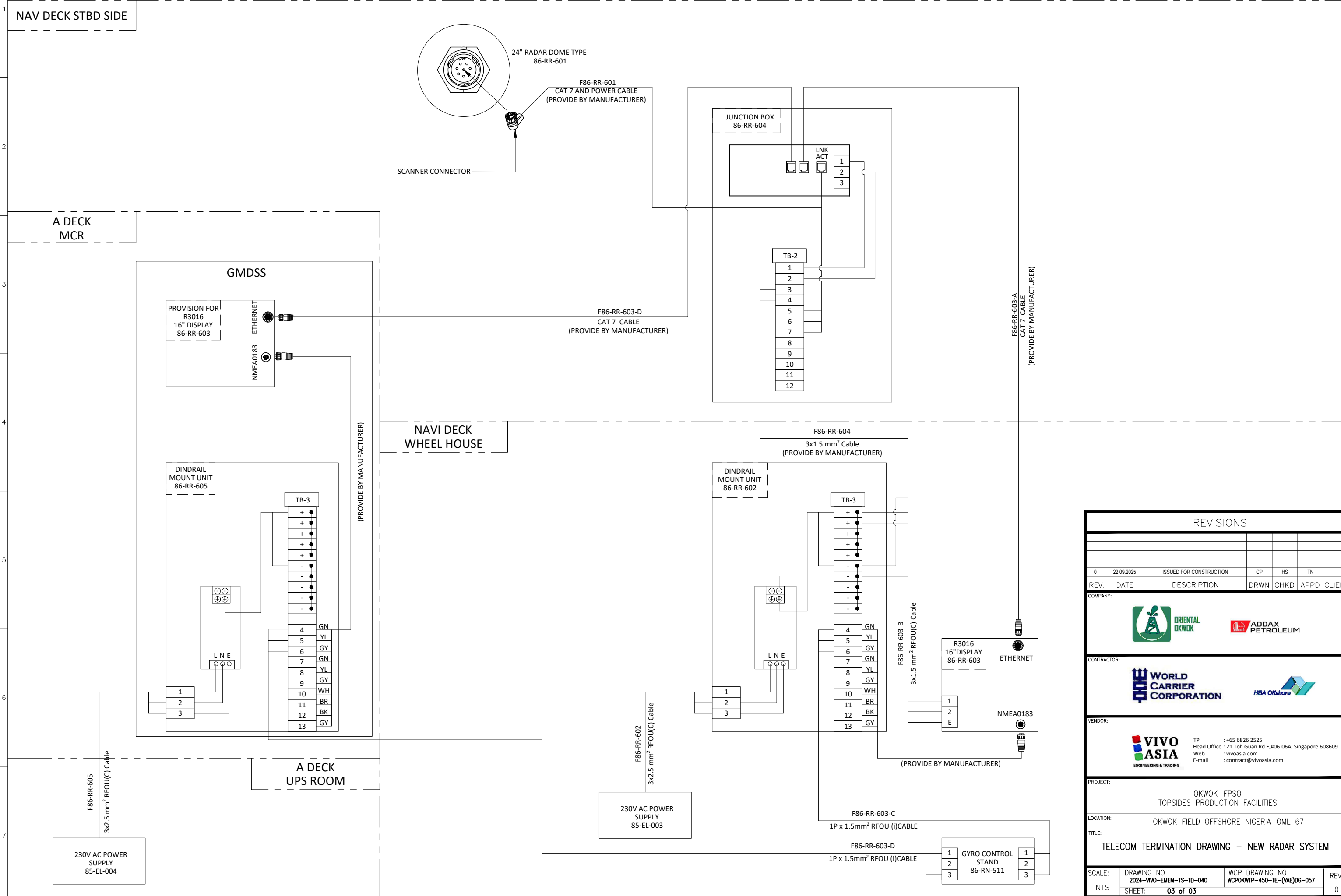
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TELECOM TERMINATION DRAWING – NEW RADAR SYSTEM

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TELECOM TERMINATION DRAWING - NEW RADAR SYSTEM



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2024-VIVO-EMEM-TS-TD-040
SHEET: 03 of 03

WCP DRAWING NO.
WCP0KWTP-450-TE-(VAE)06-057

REV.

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